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Profitability under Carbon Pricing and Exploration Risk

A Techno-Economic Comparison of Offshore Oil and Offshore Wind-Based Green Hydrogen

Group Paper

Amirhossein Jalaliyazdi

Naveed Ahmed

Supervisor: Prof. Dr. Valentin Bertsch

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Abstract

Investment continues to flow into offshore fossil fuel projects while offshore green hydrogen struggles to reach final investment decisions, despite tightening European climate policy. This paper quantifies the economic logic behind this preference. Two capital-matched projects (≈ €3 billion) are compared in an integrated discounted cash flow (DCF) and Monte Carlo framework: a stylised deepwater oil development based on the cost regressions of Qiu et al. (2022) with an EU Emissions Trading System (ETS) overlay, and a 500 MW offshore wind farm coupled to a 400 MW Proton exchange membrane (PEM) electrolyser in the German Bight. At base assumptions ($45/bbl, €5.50/kg H_2), the oil project yields a Net present value (NPV) of $502 million (Internal rate of return 12.4 %) against −€1,909 million for hydrogen, whose levelised cost of €10.96/kg far exceeds achievable offtake prices. Introducing a 30 % exploration success gate reduces oil's expected Net present value (NPV) to $71 million and its probability of profit from 91 % to 27 %-yet this remains superior to hydrogen's near-certain loss. Carbon pricing on production emissions eliminates oil profitability only above ≈ €224/t CO_2. The results indicate that the fossil investment preference is rational under current market design and persists because combustion (Scope 3) emissions are externalised; it is reversed only by hydrogen offtake support combined with low-cost capital.

Kurzfassung

Diese Arbeit vergleicht zwei kapitalgleiche Offshore-Energieprojekte (≈ 3 Mrd. €) in einem integrierten Discounted cash flow (DCF) und Monte-Carlo-Modell: eine stilisierte Tiefsee-Ölfeldentwicklung auf Basis der Kostenregressionen von Qiu et al. (2022) mit EU Emissions Trading System ETS-Kostenaufschlag sowie einen 500-MW-Offshore-Windpark mit 400-MW Proton exchange membrane PEM-Elektrolyseur in der Deutschen Bucht. Im Basisfall (45 $/bbl; 5,50 €/kg H_2) erzielt das Ölprojekt einen Kapitalwert von 502 Mio. $ (Internal rate of return 12,4 %), während das Wasserstoffprojekt mit −1.909 Mio. € deutlich negativ bleibt; die Wasserstoffgestehungskosten von 10,96 €/kg übersteigen erzielbare Abnahmepreise erheblich. Ein Explorationsrisiko-Gate (Erfolgswahrscheinlichkeit 30 %) senkt den Erwartungswert des Ölprojekts auf 71 Mio. $ und die Gewinnwahrscheinlichkeit von 91 % auf 27 % - dennoch bleibt es dem Wasserstoffprojekt überlegen. Eine CO_2-Bepreisung der Produktionsemissionen eliminiert die Ölrentabilität erst oberhalb von ca. 224 €/t. Die Investitionspräferenz für fossile Projekte erweist sich damit als ökonomisch rational, solange Verbrennungsemissionen (Scope 3) externalisiert bleiben; sie kehrt sich nur bei gestützten Wasserstoff-Abnahmepreisen in Verbindung mit niedrigen Kapitalkosten um.

Table of contents

Abstract II

Kurzfassung III

Table of contents IV

List of figures VI

List of tables VII

List of abbreviations VIII

List of symbols IX

1. Introduction 1

2. Methodology 2

2.1 Comparison framework and system boundaries 2

2.2 Offshore oil model (Model A) 2

2.3 Offshore wind-to-hydrogen model (Model B) 3

2.4 Carbon pricing, exploration risk and fiscal stress 3

2.5 Uncertainty analysis 4

3. Data and assumptions 4

4. Results and discussion 6

4.1 Base-case results 6

4.2 Sensitivity analysis 7

4.3 Carbon price scenarios and the Scope 3 boundary 8

4.4 Exploration risk and the structure of uncertainty 9

4.5 Crossover conditions 10

4.6 Limitations 11

5. Summary and conclusion 12

5.1 Summary 12

5.2 Conclusion and outlook.....12

6. References.....13

Appendix.....15

Declaration on the use of generative AI……..…16

List of figures

Figure 1: Annual net cash flow profiles of both projects (base case).....6

Figure 2: Monte Carlo NPV distributions (3,000 runs; oil conditional on discovery).....7

Figure 3: Tornado diagrams of NPV sensitivity (parameters sorted by swing).....8

Figure 4: Oil NPV versus EU ETS price; Scope 3 stress test indicated.....9

Figure 5: Cumulative NPV distribution of the oil project with and without the exploration gate..... 10

Figure 6: Hydrogen NPV (€M) across offtake price and WACC; contours mark NPV = 0 and parity with the oil base case.....11

List of tables

Table 1: Key assumptions, offshore oil (Model A)..... 4

Table 2: Key assumptions, wind-to-hydrogen (Model B, PEM)..... 5

Table 3: Base-case results and Monte Carlo risk metrics..... 6

List of abbreviations

Abbreviation	Explanation
ALK	Alkaline electrolysis
CAPEX	Capital expenditure
CfD	Contract for Difference
DCF	Discounted cash flow
E&A	Exploration and appraisal
ETS	(EU) Emissions Trading System
FPSO	Floating production, storage and offloading unit
IRR	Internal rate of return
LCOH	Levelised cost of hydrogen
MC	Monte Carlo (simulation)
NPV	Net present value
OPEX	Operating expenditure
PEM	Proton exchange membrane (electrolysis)
PoS	Probability of (geological/commercial) success
WACC	Weighted average cost of capital

List of symbols

Symbol	Unit	Description
	Mbbl	Original oil in place
	Mbbl	Recoverable reserves
	km²	Drainage area
	m	Net pay thickness
	–	Porosity
	–	Oil saturation
	–	Formation volume factor
	–	Recovery factor
	$M / €M	Net cash flow in year
	–	Discount rate (WACC)
	a	Project lifetime

1. Introduction

Global upstream oil and gas investment has remained above 500 billion USD per year even as the European Union has tightened its climate framework, while offshore green hydrogen-widely presented as the successor technology for the North Sea energy system-has seen only a handful of final investment decisions. This apparent contradiction motivates the present study: if carbon pricing is supposed to redirect capital, why does capital still prefer projects with substantial geological realisation risk over a technology with no resource risk at all?

The research question of this paper is: Accounting for exploration risk and cost uncertainty, how do the profitability indicators of offshore oil projects compare to offshore wind-based green hydrogen projects across carbon price scenarios, and what drives the observable investment preference for fossil fuels despite higher realisation risk?

Existing technoeconomic literature treats the two asset classes largely in isolation. For offshore oil, Qiu et al. (2022) provide regression-based cost models for deepwater developments together with a Monte Carlo assessment of geological uncertainty. For offshore hydrogen, studies such as Singlitico et al. (2021), Travaglini et al. (2024) and Glaum et al. (2024) estimate levelised costs of hydrogen (LCOH) for wind-electrolyser systems, generally finding values well above current willingness to pay. Direct, capital-matched comparisons of the two investment alternatives under a common carbon-policy overlay-and with exploration risk made explicit-are rare. This paper addresses that gap.

The contribution is threefold. First, two full project models are integrated in a single framework and matched on invested capital (≈ €3 billion), which is the comparison basis an investor allocating a fixed budget actually faces. Second, uncertainty is treated symmetrically where possible (Monte Carlo simulation with triangular distributions on both sides) and asymmetrically where reality is asymmetric: a Bernoulli exploration gate applies only to oil, because only oil carries resource risk. Third, carbon liability is examined at two system boundaries-production emissions under the current EU ETS, and an illustrative full-lifecycle (Scope 3) stress test-to show how the choice of boundaries, rather than the carbon price level, determines the outcome.

The remainder of the paper is organised as follows. Chapter 2 develops the methodology and system boundaries. Chapter 3 summarises data and key

assumptions. Chapter 4 presents base-case results, sensitivity and scenario analyses, the effect of the exploration gate, and the conditions under which the ranking reverses, followed by a discussion of limitations. Chapter 5 concludes.

2. Methodology

2.1 Comparison framework and system boundaries

Three matching bases were considered: equal energy output, equal revenue, and equal capital employed. Energy-equivalent scaling would require more than 12 GW of offshore wind to match the field's ≈ 88 PJ/a output and is irrelevant to a capital-allocation decision; revenue matching suffers from the same objection. The comparison is therefore capital-matched (Framework 3): total CAPEX of $3,067 million for the oil development versus €2,938 million for the wind-hydrogen system, a difference of about 4 %, treated at EUR/USD parity of 1.0 throughout.

The oil model is an archetype, not a site-specific case: a stylised deepwater development (1,200 m water depth) parameterised on the only peer-reviewed offshore cost regression set available (Qiu et al., 2022, calibrated on Bohai Bay data), evaluated under an EU ETS policy overlay and the fiscal terms of that framework (10 % royalty, 11 % income tax). A literal North Sea field would sit in shallow water and face petroleum taxation of 60–78 %; the consequences of this jurisdictional mismatch are quantified explicitly in Section 4.5 rather than hidden in the base case. The hydrogen project is in the German Bight (average 58 km to shore; Fraunhofer IEE, 2024). Both projects run over an identical 25year horizon. Discount rates are risk-adjusted per technology-10 % for oil and 7 % (based) for hydrogen, the latter additionally sampled between 3 % and 15 % in the Monte Carlo analysis to reflect the wide observed financing spectrum for green hydrogen (OECD, 2024). System boundaries were deliberately symmetrised during model development: both projects carry decommissioning obligations (10 % of CAPEX for oil in year 25; a 3 % reserve for the wind-hydrogen system), and the electrolyser production accounts for the physical clipping of wind output above its rating.

2.2 Offshore Oil Model (Model A)

Recoverable reserves follow the volumetric method,

yielding 378 Mbbl at base parameters (original oil in place 1,145 Mbbl). Production follows a plateau profile: two construction years, ramp-up (30 %/65 % of the 55,000 bbl/d plateau), plateau until year 15, then 15 %/a decline, delivering 331 Mbbl over the horizon. CAPEX comprises four regression-based components-drilling, underwater systems, substructure and topsides/FPSO (Qiu et al., 2022, Eqs. 5–8)-phased 60/40 over years 1–2; OPEX follows the corresponding per-barrel regression. Revenue includes associated gas (gasoil ratio 87.6 m^3/m^3 at \$0.06/m^3). The EU ETS liability prices production emissions of 17 kg CO_2/bbl (IEA, 2023) at €65/t (2024 average; European Commission, 2025), escalating at 7 %/a. Net cash flow in year is revenue less royalty, CAPEX, OPEX, ETS cost, decommissioning and income tax; the NPV is

2.3 Offshore wind-to-hydrogen model (Model B)

A 500 MW fixed-bottom wind farm (net capacity factor 46 %; IRENA, 2024; NREL, 2024) feeds a 400 MW PEM electrolyser (system CAPEX €2,800/kW; IEA, 2024), sized at the standard 0.8 ratio (Singlitico et al., 2021). Because a 400 MW electrolyser cannot absorb wind output above its rating, a 5 % clipping loss is applied to hydrogen production. Annual net output is 33,976 t H_2 after 3.5 % compression losses, transported through an 80 km subsea pipeline (€0.50/kW_{H2}/km; Glaum et al. 2024). CAPEX (wind €2.9M/MW, electrolyser, pipeline and compression) carries a 1.12 overrun factor on the wind and electrolyser components (Merrow, 2011) and is phased 40/40/20 over three construction years. OPEX comprises wind (€65k/MW/a), electrolyser (3.5 % of CAPEX/a) and pipeline components; a stack replacement (45 % of stack cost) falls in year 21 and a decommissioning reserve in year 25. Hydrogen sells at €5.50/kg (IEA, 2022; Travaglini et al., 2024); corporate tax is 25 %. The levelised cost of hydrogen is defined as

2.4 Carbon pricing, exploration risk and fiscal stress

Three policy-risk mechanisms extend the base models. First, the ETS price is varied between €0 and €400/t, with €120/t as the 2030 central scenario (BNEF, 2024). Second, an illustrative Scope 3 stress test asks what the oil NPV would be if the producer internalised full lifecycle emissions of 450 kg CO_2/bbl instead of 17 kg; this is an accounting exercise on carbon liability, not a prediction of regulation, since

combustion emissions are priced at the point of combustion in the EU system and a producer-side charge of this size would shift market prices. Third, exploration risk enters through a Bernoulli gate: with probability of success = 30 % (Westwood Global Energy, 2024; Norwegian Offshore Directorate, 2024) the field is discovered and developed; otherwise only the sunk exploration and appraisal cost = $80 million is lost. The unconditional expected NPV is

This formulation conservatively ignores the option to abandon uneconomic discoveries after appraisal, and therefore understates oil value. Finally, a fiscal stress scenario reprices the oil project's tax burden at a North-Sea-style 65 % petroleum tax by linear scaling of the tax present value, addressing the jurisdictional mismatch of the archetype directly.

2.5 Uncertainty Analysis

Both models undergo 1,000 run Monte Carlo simulations with triangular distributions. For oil, seven parameters are sampled (recovery factor, oil price, porosity, net pay, formation volume factor, water depth, pipeline diameter), with ranges and variance contributions anchored to Qiu et al. (2022); the simulation uses present-value factors derived from the DCF and approximates revenue by recoverable reserves times price, which places its central value above the DCF-a known and disclosed property. For hydrogen, nine parameters are sampled, including CAPEX components, capacity factor, efficiency, offtake price, WACC, pipeline length, overrun and an OPEX multiplier. One-at-a-time sensitivity analyses (tornado diagrams) complement the simulations. All computations are implemented in a fully formula-based Excel workbook (63,746 formulas) accompanying this paper; every figure reported here is reproducible from it.

3. Data and assumptions

Table 1 and Table 2 summarise the key inputs; the complete, source-annotated assumption sets are contained in the workbook (sheets Assumptions_Oil and Assumptions_H2).

Table 1: Key assumptions, offshore oil (Model A)

Parameter	Base	Range (MC)	Source
Recovery factor	0.33	0.11–0.64	Qiu et al. (2022)

Parameter	Base	Range (MC)	Source
Oil price (€/bbl)	45	30–70	Qiu et al. (2022)
Total CAPEX (€M)	3,067	regression	Qiu et al. (2022)
WACC	10 %	fixed	Qiu et al. (2022)
ETS price (€/t), escal. 7 %/a	65	50–200	European Commission (2025)
CO_2 intensity (kg/bbl)	17	10–28	IEA (2023)
PoS / E&A cost (€M)	0.30 / 80	0.20–0.40	Westwood (2024)
Decommissioning (% of CAPEX)	10 %	5–15 %	OSPAR/NSTA guidance

Table 2: Key assumptions, wind-to-hydrogen (Model B, PEM)

Parameter	Base	Range (MC)	Source
Wind capacity / CAPEX	500 MW / €2.9M/MW	2.4–4.1	IRENA (2024)
Capacity factor	0.46	0.38–0.54	IRENA (2024); NREL (2024)
Electrolyser CAPEX (€/kW)	2,800	1,000–3,500	IEA (2024)
Efficiency (kWh/kg)	52	48–64	DOE (2024); IEA (2024)
Clipping loss	5 %	fixed	Singlitico et al. (2021)
H_2 price (€/kg)	5.50	2.5–10	IEA (2022); Travaglini et al. (2024)
WACC	7 %	3–15 %	OECD (2024)
CAPEX overrun factor	1.12	1.0–1.5	Merrow (2011)

The resulting cashflow structures are shown in Figure 1: oil is characterised by two large negative construction years followed by high, declining surpluses; hydrogen by three construction years followed by 22 small, flat surpluses interrupted by the stack replacement.

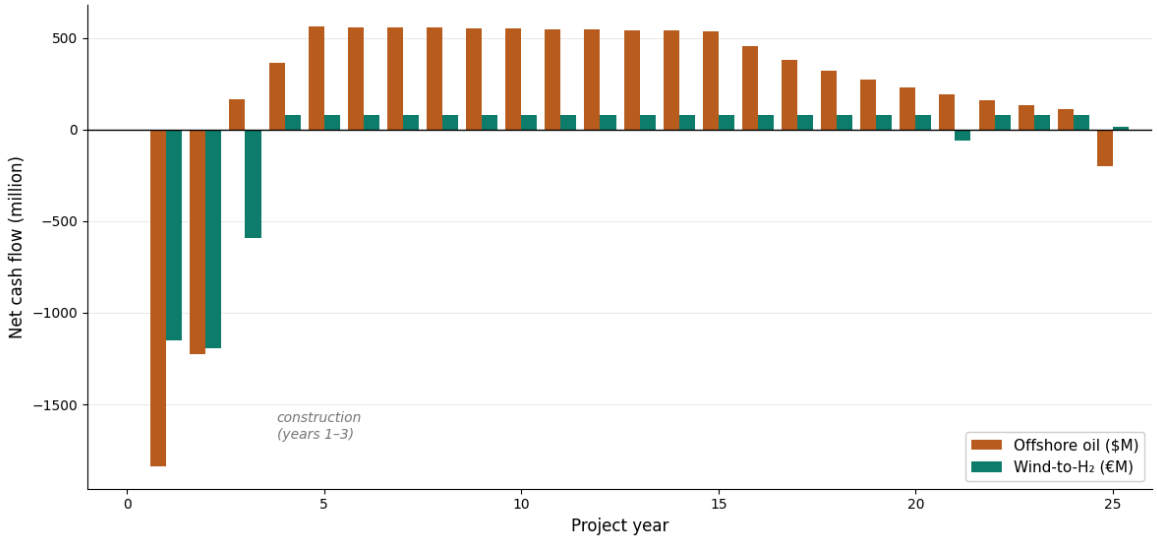


Figure 1: Annual net cash flow profiles of both projects (base case)

4. Results and discussion

4.1 Base-case results

Table 3 reports the base case. The oil development is comfortably profitable: NPV of €502 million at 10 % WACC, IRR of 12.4 %, and a breakeven oil price of €39.7/bbl including carbon and decommissioning costs (€37.5/bbl without the ETS). The hydrogen project is deeply unprofitable at market conditions: NPV of −€1,909 million and an IRR of −4.8 %. The economic diagnosis is direct: the levelised cost of hydrogen of €10.96/kg is roughly twice the assumed offtake price of €5.50/kg. Because of the tax asymmetry-tax is levied on profitable operating years while construction outlays receive no relief-the after-tax NPV reaches zero only at ≈ €13.8/kg.

Table 3: Base-case results and Monte Carlo risk metrics

Metric	Offshore oil	Wind to H_2 (PEM)
Total CAPEX	€3,067M	€2,938M
NPV (base)	€502M	−€1,909M
IRR	12.4 %	−4.8 %
Breakeven / LCOH	€39.7/bbl	€10.96/kg
MC mean NPV (3,000 runs)	€2,682M [1]	−€1,855M
P(NPV > 0), conditional	91 %	1 %
P(NPV > 0), unconditional	27 %	1 % (no resource risk)
E[NPV] incl. exploration gate	€71M	−€1,909M

[1] Simulation value based on reserves × price approximation; lies structurally above the DCF value (Section 2.5).

The Monte Carlo distributions (Figure 2) reveal the deeper difference: the two projects do not merely have different expected values; they have different shapes of risk. Oil's distribution is wide and right-skewed-driven by the recovery factor and oil price-with 91 % of runs profitable conditional on discovery. Hydrogen's distribution is narrower but lies almost entirely below zero; even the 95th percentile (−€719M) does not reach profitability. Uncertainty is therefore not the hydrogen project's problem; the central estimate is.

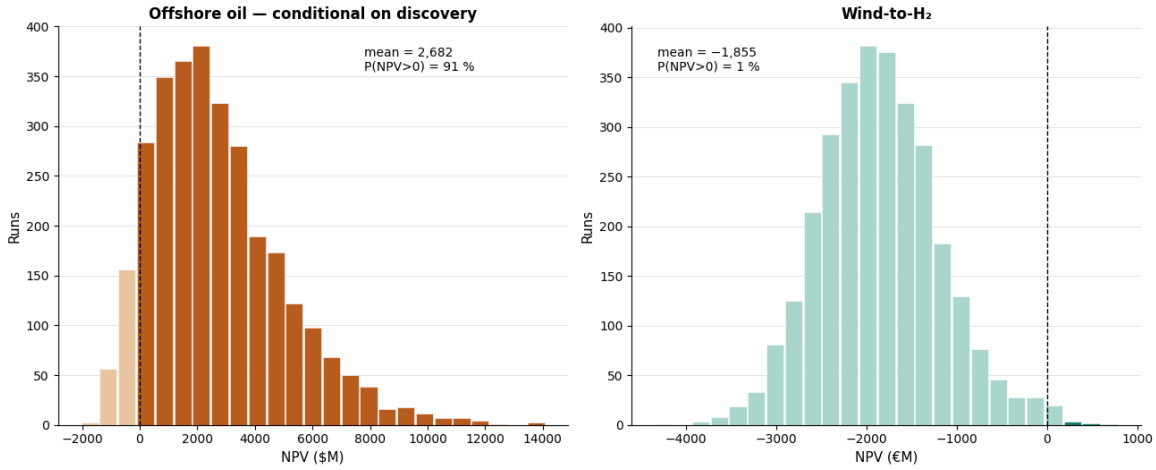


Figure 2: Monte Carlo NPV distributions (3,000 runs; oil conditional on discovery)

4.2 Sensitivity analysis

The tornado diagrams (Figure 3) identify the value drivers. For oil, the geological recovery factor dominates (swing ≈ €6.1 billion), followed by the oil price (≈ €4.3 billion)-reproducing the ranking of Qiu et al. (2022) and confirming that subsurface uncertainty, not policy, is the primary risk of a fossil project. The EU ETS price, even across €50–200/t, produces a swing of under $0.5 billion, an order of magnitude smaller. For hydrogen, the offtake price dominates (swing ≈ €2.3 billion), followed by electrolyser CAPEX, the CAPEX overrun factor, the capacity factor and WACC. The two projects are thus exposed to fundamentally different markets: oil to the resource and the commodity price, hydrogen to a policy-constructed offtake market and to the cost of capital.

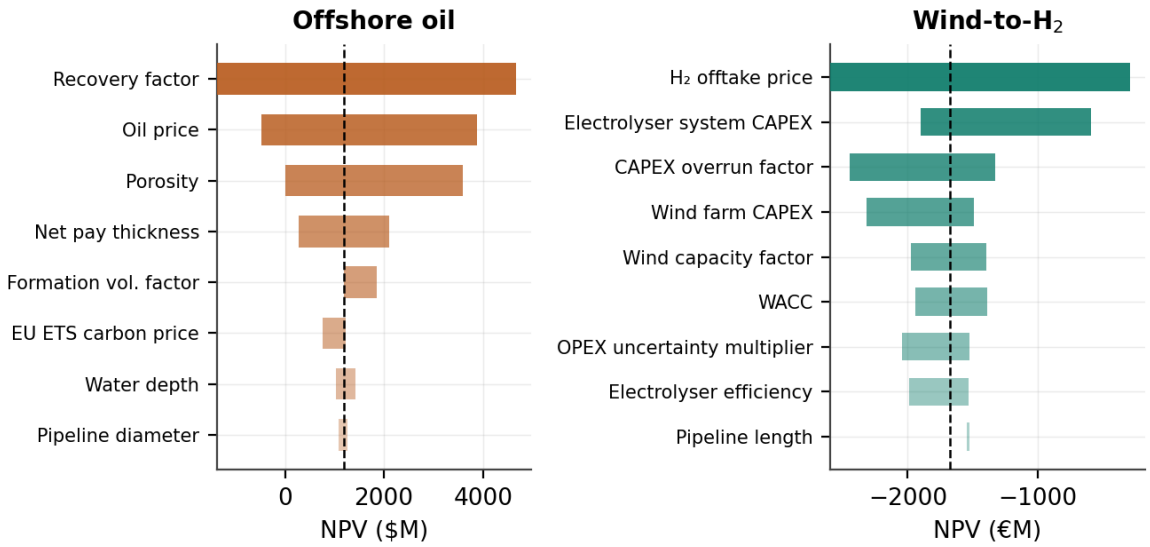


Figure 3: Tornado diagrams of NPV sensitivity (parameters sorted by swing)

4.3 Carbon price scenarios and the Scope 3 boundary

Figure 4 traces oil NPV across ETS prices from €0 to €400/t. At the €65/t base, the ETS costs the project €206 million of NPV (€708M → €502M)-material, but far from decisive. Oil NPV reaches zero only at ≈ €224/t (base year equivalent, including 7 %/a escalation), well above any politically plausible trajectory for the 2020s; the 2030 central scenario of €120/t still leaves ≈ $330 million of NPV. Carbon pricing of *production* emissions therefore erodes, but cannot eliminate, the fossil advantage.

The system boundary, not the price, is decisive. If the producer internalised the full lifecycle intensity of 450 kg CO_2/bbl, the NPV would collapse to −€5,378 million at €65/t and −€10,549 million at €120/t. This stress test must be read as carbon-liability accounting rather than a policy scenario-combustion emissions are priced at combustion in the EU, and a supply-side charge of this size would shift market prices via the supply curve-but it exposes the mechanism precisely: the investment preference for oil exists because roughly 96 % of its lifecycle emissions sit outside the producer's balance sheet.

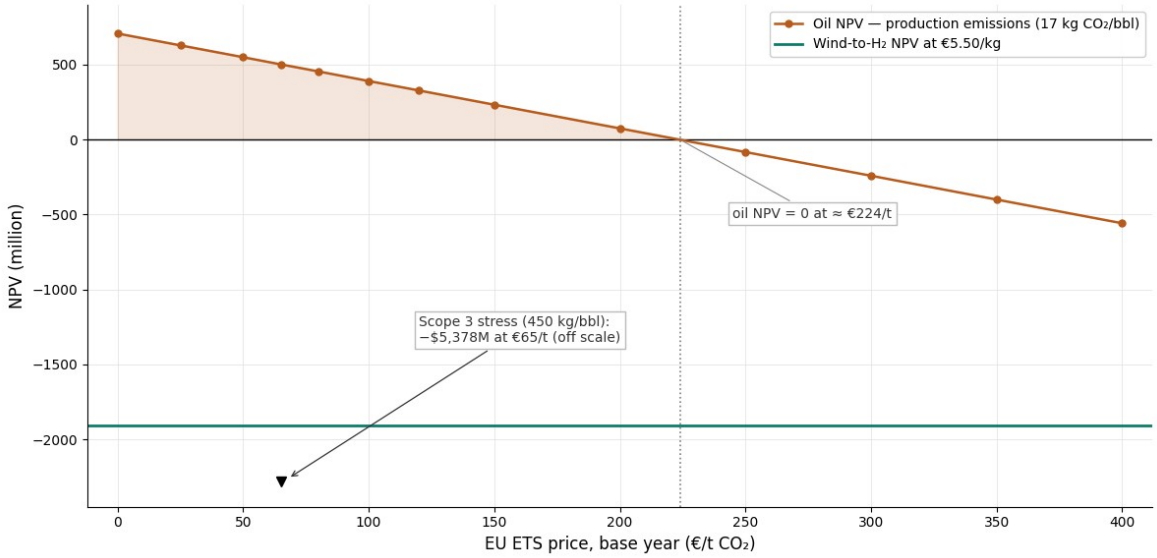


Figure 4: Oil NPV versus EU ETS price; Scope 3 stress test indicated

4.4 Exploration risk and the structure of uncertainty

Introducing the exploration gate transforms the oil investment case (Figure 5). Unconditionally, the probability of a positive NPV falls from 91 % to 27 %, and 70 % of outcomes are a total loss of the €80 million E&A budget; the expected NPV drops from €502 million to €71 million. Yet even this heavily risked figure exceeds hydrogen's certain −€1,909 million by two orders of magnitude in expectation.

This resolves the apparent paradox in the research question. The fossil preference does not persist despite higher realisation risk; it persists because realisation risk of the oil type is economically tolerable: a bounded, cheap-to-resolve lottery ticket ($80M buys a 30 % chance at a multibillion upside), diversifiable across an exploration portfolio. Hydrogen offers the mirror image: no resource risk, full technical certainty of delivery-and a near-certain economic loss under market prices. A risk-neutral, and even a moderately risk-averse, investor rationally prefers the lottery. The comparison of the two probability metrics must nonetheless be made carefully: oil's 27 % reflects a discrete geological gamble, hydrogen's 1 % a continuous price-cost distribution; they describe different risk structures, not merely different levels.

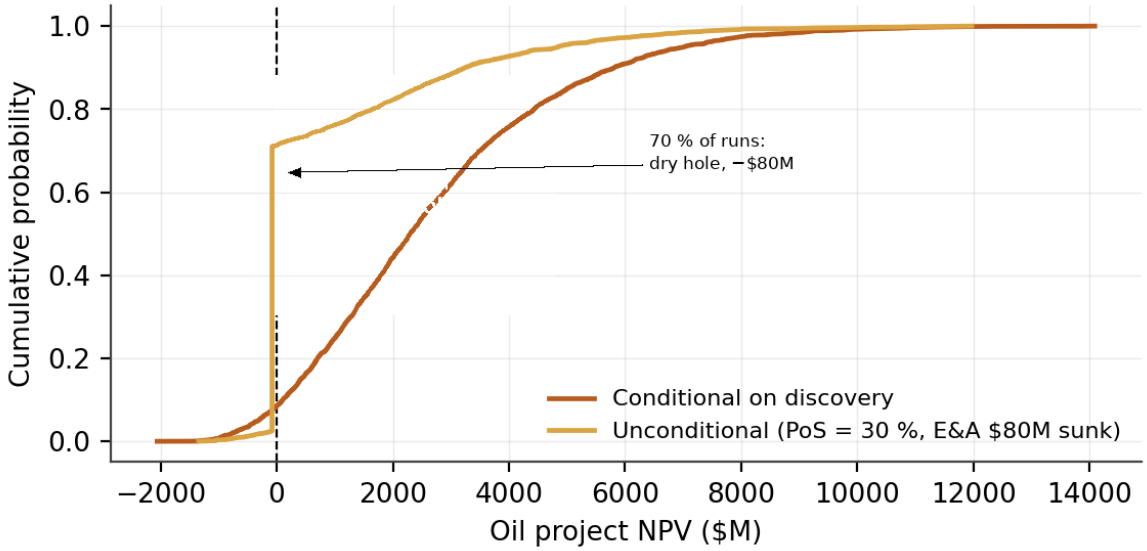


Figure 5: Cumulative NPV distribution of the oil project with and without the exploration gate

4.5 Crossover conditions

Under which conditions does the ranking reverse? Figure 6 maps hydrogen NPV across offtake price and WACC. Hydrogen reaches standalone viability (after tax) only above ≈ €10–14/kg depending on financing costs, and beats the oil base case only in the upper-left region-prices near €14–16/kg combined with WACC below ≈ 5 %. Named scenarios in the accompanying workbook confirm the pattern: neither an equal WACC case, nor €300/t carbon, nor low oil prices alone suffice; only combined scenarios (“Full Green Transition”: €35/bbl, €12/kg, 3.5 % WACC, €300/t) flip the decision. Under a North-Sea-style 65 % petroleum tax, however, the oil archetype itself turns deeply negative (−€1,444M)-indicating that fiscal design rivals carbon pricing as a lever over fossil investment, and that part of the modelled oil advantage stems from the mild fiscal terms of the underlying cost framework.

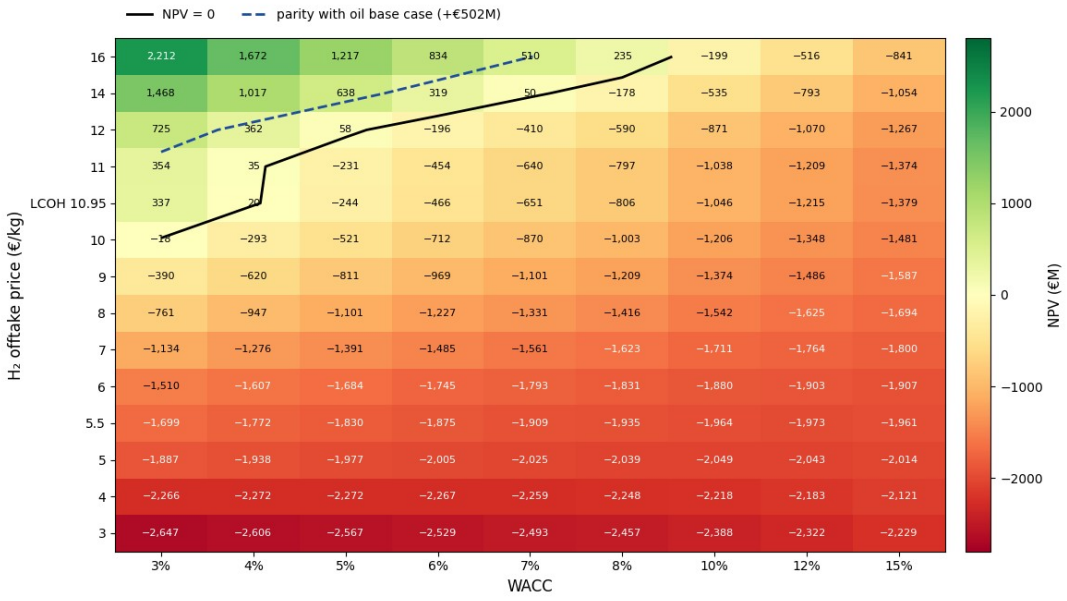


Figure 6: Hydrogen NPV (€M) across offtake price and WACC; contours mark NPV = 0 and parity with the oil base case

4.6 Limitations

Five limitations qualify these results; their directions are known and none overturns the central finding. First, the oil model is a jurisdictional archetype: Bohai-calibrated deepwater costs and mild fiscal terms under an EU carbon overlay. A genuinely European project would face far higher taxation (quantified above: NPV −€1,444M at 65 %) but also shallower water and lower CAPEX; the base case therefore brackets rather than represents a real field. Second, the exploration gate ignores the post-appraisal abandonment option and compresses exploration timing into year 0, both of which *understate* oil value-the fossil-preference conclusion is thus conservative. Third, the Scope 3 stress abstracts from price incidence and from double-counting with demand-side pricing; it quantifies liability exposure, not equilibrium outcomes. Fourth, the hydrogen model excludes the policy support stack (H2Globaltype premia, CfDs, grants) that real projects depend on-the upper-left region of Figure 6 is precisely the space such instruments target-and it does not test the wind-to-grid counterfactual, under which the electricity consumed per kilogram of hydrogen (52 kWh) would likely earn more sold directly. Fifth, technical simplifications remain: EUR/USD parity, uncorrelated Monte Carlo parameters, flat real commodity prices against a nominally escalating ETS, and a PV-factor-based simulation on the oil side. Sensitivity of the conclusions to each was examined in the workbook; the qualitative ranking survived every variation tested.

5. Summary and conclusion

5.1 Summary

This paper compared a capital-matched offshore oil development and an offshore wind-to-hydrogen project (≈ €3 billion each) in an integrated DCF and Monte Carlo framework with an EU ETS overlay, an explicit exploration gate, and symmetric system boundaries. At base conditions the oil project earns €502 million NPV (IRR 12.4 %) while the hydrogen project loses €1,909 million, its LCOH of €10.96/kg standing at roughly twice the achievable offtake price. Carbon pricing of production emissions reverses the oil result only above ≈ €224/t CO_2. The exploration gate cuts oil's success probability to 27 % and its expected NPV to €71 million-still strictly superior to hydrogen's near-certain loss. The ranking reverses only under combined conditions: hydrogen offtake prices near €14/kg together with capital costs below ≈ 5 %, or European-style petroleum taxation of the oil side.

5.2 Conclusion and outlook

The answer to the research question is uncomfortable but clear: the observable investment preference for offshore fossil projects is rational under current market design. It rests on three pillars-the externalisation of combustion emissions (≈ 96 % of lifecycle carbon lies outside the producer's ETS liability), the tolerability of geological risk as a cheap, bounded, diversifiable lottery, and the absence of a market-priced hydrogen offtake at cost-covering levels. It follows that carbon prices within any politically plausible range will not redirect this capital on their own. The effective levers lie elsewhere: credible long-term hydrogen offtake instruments that lift effective prices toward LCOH, derisking mechanisms that push financing costs toward the 3–5 % range, and fiscal design on the fossil side. Future work should endogenise the wind-to-grid counterfactual, model the hydrogen support stack explicitly, and replace the archetype oil fiscal regime with a consistent European one-extensions for which the accompanying open workbook provides a directly adaptable basis.

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Appendix

The complete model accompanies this paper as an Excel workbook (Integrated_Offshore_Energy_Model.xlsx, 13 sheets, 63,746 formulas). Sheet README documents structure, methodology, the audit log and known limitations. All assumptions carry source annotations (Assumptions_Oil, Assumptions_H2); the DCF sheets, Monte Carlo simulations (F9 rerolls), sensitivity tables and all scenario analyses reported in Chapter 4 are fully formula-linked and reproducible. Static scenario tables and the 3,000run histogram were regenerated from the final model version; consistency between live and static values was verified programmatically at all anchor points.

Declaration on the use of generative AI

During the preparation of this paper, I used the following AI tools for the purposes described below. After using these tools, I reviewed and edited the content as needed and take full responsibility for the content of the paper.

Type of use	Scope of use	Name of the AI tool used	Name of the AI tool used
Ideation and conceptualization	e.g., development of ideas, objectives, and research questions; identification and definition of relevant concepts	ChatGPT, Gemini, Claude	entire work
Literature search and analysis	e.g., search for relevant literature; review of potentially relevant literature; summaries of relevant literature	ChatGPT, Gemini, Claude	entire work
Methodological approach	e.g., search for a suitable methodological approach; development and adaptation of the methodological approach to the research question(s)	ChatGPT, Gemini, Claude	entire work
Data collection and analysis	e.g. qualitative data analysis including transcription and coding; quantitative data analysis including statistical evaluation	ChatGPT, Gemini, Claude	entire work
Presentation and Discussion of Results	e.g., processing and presentation of the results; interpretation and discussion of the results in the context of the state of research; derivation of implications for research and practice	ChatGPT, Gemini, Claude	entire work
Text processing, e.g., (i) text structuring, (ii) text generation, (iii) text translation, (iv) text review and correction	re (i) e.g., structuring the entire paper and/or individual chapters or sections re (ii) e.g., generating text on various topics across different chapters, including titles and abstracts re (iii) e.g., translating one's own text or texts by other authors re (iv) e.g., reviewing and revising—or correcting—the content, organization, or grammar of the paper; rephrasing or paraphrasing text; shortening or expanding the text	ChatGPT, Gemini, Claude	entire work
Source management	e.g., compiling the bibliography; maintaining and documenting sources	ChatGPT, Gemini, Claude	entire work